Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The function $f(x) = \frac{x^2}{e^x}$ monotonically increasing if

(A) $x < 0$ only

(B) $x > 2$ only

(C) $x \in (0, 2)$

(D) $x \in (-\infty, 0) \cup (2, \infty)$

Q.2

If f and g are two increasing functions such that fog is defined, then which one of the following is correct?

(A) fog is always an increasing function

(B) fog is always a decreasing function

(C) fog is neither an increasing nor a decreasing function

(D) None of the above

Q.3

If $f: A \to B$ where $A = \{2, 3, 4\}$ and $B = \{6, 2, 3, 4, 5\}$,then the number of non-decreasing functions on f that can be defined are ____

(A) 30

(B) 35

(C) 40

(D) 45

Q.4

The function $f(x) = 2\log(x - 2) - x^2 + 4x + 1$ increases on the interval

(A) $(1, 2)$

(B) $(2, 3)$

(C) $(1, 4)$

(D) $(2, 4)$

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Q.5

Consider the following statements: 1.$\mathrm{f(x) = \ln x}$ is an increasing function on $(0, \infty)$. 2.$\mathrm{f(x) = e^x - x(\ln x)}$ is an increasing function on $(1, \infty)$. Which of the above statements is/are correct?

(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.6

What is the interval in which the function $f(x) = \sqrt{9 - x^2}$ is increasing? $(f(x) > 0)$

(A) $(0, 3)$

(B) $(-3, 0)$

(C) $[-3, 0)]$

(D) $[-3, 3]$

Q.7

Let f and g be functions from the interval $[0, \infty)$ to the interval $[0, \infty)$, f being an increasing and g being a decreasing function. If $\mathrm{f\{g(0)\} = 0}$ then

(A) $\mathrm{f\{g(x)\} \geq f\{g(0)\}}$

(B) $g\{f(x)\} \leq g\{f(0)\}$

(C) $\mathrm{f\{g(2)\} = 7}$

(D) None of these

Q.8

The range of the function $f(x) = 2\sqrt{x - 2} + \sqrt{4 - x}$ is

(A) $(\sqrt{2}, \sqrt{10})$

(B) $[\sqrt{2}, \sqrt{10})$

(C) $(\sqrt{2}, \sqrt{10}]$

(D) $[\sqrt{2}, \sqrt{10}]$

Q.9

$f(x) = \frac{\log(\pi + x)}{\log(e + x)}$ is

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(A) increasing in $[0, \infty)$

(B) decreasing in $[0, \infty)$

(C) decreasing in $\left[0, \frac{\pi}{e}\right]$ & increasing in $\left[\frac{\pi}{e}, \infty\right)$

(D) increasing in $\left[0, \frac{\pi}{e}\right]$ & decreasing in $\left[\frac{\pi}{e}, \infty\right)$

Q.10

If $f(x) = kx^3 - 9x^2 + 9x + 3$ is monotonically increasing in every interval, then which one of the following is correct?

(A) $k < 3$

(B) $k \leq 3$

(C) $k > 3$

(D) $k \geq 3$

Q.11

A function $g(x)$ is defined as $g(x) = \frac{1}{4} f\left(2x^2 - 1\right) + \frac{1}{2} f\left(1 - x^2\right)$ and $f'(x)$ is an increasing function. Then $g(x)$ is increasing in the interval

(A) $(-1, 1)$

(B) $\left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$

(C) $\left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right)$

(D) None of these

Q.12

If $\frac{x^2}{f4a} + \frac{y^2}{fa^2 - 5}$ represents an ellipse with major axis as y-axis and f is a decreasing function, then:

(A) $a \in (-\infty, 1)$

(B) $a \in (5, \infty)$

(C) $a \in (1, 4)$

(D) $a \in (-1, 5)$

Q.13

The function $f : [0, 3] \to [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

(A) one-one and onto

(B) onto but not one-one

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(C) one-one but not onto

(D) neither one-one nor onto

Q.14

Let $fx = 2x + \cot^{-1}x + \ln\sqrt{1+x^2}$ - x. Then fx

(A) increases in $(-\infty, \infty)$

(B) decreases in $(-\infty, \infty)$

(C) neither increases nor decreases in $(0, \infty)$

(D) increases as well as decreases in $(-\infty, \infty)$

Q.15

$f(x) = 2x^2 - \log|x| (x \neq 0)$ is monotonic increasing in the interval

(A) $(1/2, \infty)$

(B) $(-\infty, -1/2) \cup (1/2, \infty)$

(C) $(-\infty, -1/2) \cup (0, 1/2)$

(D) $(-1/2, 0) \cup (1/2, \infty)$

Q.16

The set of all values of a for which the function $f(x) = \left(a^2 - 3a + 2\right)\left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4}\right) + (a - 1)x + \sin 1$ does not possess critical points is

(A) $[1, \infty)$

(B) $(0, 1) \cup (1, 4)$

(C) $(-2, 4)$

(D) $(1, 3) \cup (3, 5)$

Q.17

Let $F(x) = x^3 + ax^2 + bx + 5\sin^2 x$ be an increasing function in the set of real number $\mathbf{R}$. Then a and b satisfy the condition.

(A) $a^2 - 3b - 15 > 0$

(B) $a^2 - 3b + 15 < 0$

(C) $a^2 + 3b + 15 < 0$

(D) $\mathrm{a} > 0$ and $\mathrm{b} > 0$

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Q.18

The function $\mathrm{f(x)} = 3\cos^4 \mathrm{x} + 10\cos^3 \mathrm{x} + 6\cos^2 \mathrm{x} - 3\ (0 \le x \le \pi)$ is $-$

(A) Increasing in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

(B) Increasing in $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$

(C) Decreasing in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

(D) All of above

Q.19

If $f(x) = \left(\frac{a^2-1}{a^2+1}\right) x^3 - 3x + 5$ is a decreasing function of x in R then the set of possible values of a (independent of x) is

(A) $(1, \infty)$

(B) $(-\infty, -1)$

(C) $[-1, 1]$

(D) None of these

Q.20

Let $f: R \to R$ be a function defined by $fx = \text{-}\,x^3 \text{-}\, 3x^2 \text{-}\, 6x + 1$. Number of integers in the solution set of x satisfying the inequality $ffx^3 + fx \ge ff\text{-}fx \text{-}\, x^3$ is

(A) 3

(B) 4

(C) 5

(D) 6

Answers & Solutions

Q.1 Answer:

$x \in (0, 2)$

Solution:

$f(x) = \frac{x^2}{e^x}; f'(x) = \frac{2x \cdot e^x - e^x \cdot x^2}{(e^x)^2}$

$f'(x) = \frac{2x - x^2}{e^x}$

as $\mathrm{e^x}$ is always positive and for monotonically increasing;

$2x - x^2 > 0 \Rightarrow \quad x^2 - 2x < 0 \Rightarrow x(x-2) < 0 \Rightarrow x \in (0, 2)$

Q.2 Answer:

fog is always an increasing function

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Solution:

Product of two increasing function is always an increasing function.

$\therefore$ fog **is always an increasing function**

Q.3 Answer:

35

Solution:

For a function to be non-decreasing,

$x_1 < x_2 \Rightarrow fx_1 \leq fx_2$

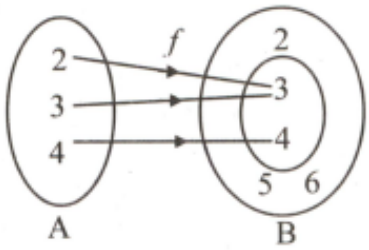
Case I : The range of f **has only one element.**

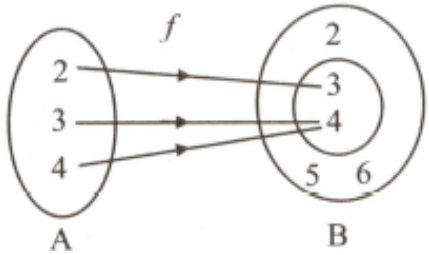
Number of functions $= 5$

Case II : The range of f **has exactly two elements.**

Select two elements from $6, 2, 3, 4, 5$ **This can be done in** ${}^5C_2$**ways.**

Say, $3, 4$ **are selected. Then there can be only two non-decreasing functions.**





$\Rightarrow$ **For every selection of 2 elements in the range, there are 2 functions**

$\Rightarrow$ **Number of functions, range of which having 2 elements** $= {}^5C_2 \times 2 = 20$

Case III : The range of function f **having** 3 **elements**

3 **elements out of five elements can be selected in** ${}^5C_3 = 10$**ways.**

Say $2, 4, 6$ **are selected then**

$f(2) = 2, \quad f(3) = 4, \quad f(4) = 6$ **is the only nondecreasing function.**

$\Rightarrow$ **For every selection of three elements, there is only one non-decreasing function.**

$\Rightarrow$ **Number of non-decreasing functions range of which having three elements** $= 10$

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$\Rightarrow$ **The required number** $= 5 + 20 + 10 = 35$

Q.4 Answer:

$(2, 3)$

Solution:

$$f(x) = 2\log(x-2) - x^2 + 4x + 1$$

$$\Rightarrow f'(x) = \frac{2}{x-2} - 2x + 4$$

$$\Rightarrow f'(x) = 2\left[\frac{1-(x-2)^2}{x-2}\right] = -2\frac{(x-1)(x-3)}{x-2}$$

$$\Rightarrow f'(x) = \frac{-2(x-1)(x-3)(x-2)}{(x-2)^2}$$

$$\therefore f'(x) > 0 \Rightarrow -2(x-1)(x-3)(x-2) > 0$$

$$\Rightarrow (x-1)(x-2)(x-3) < 0 \Rightarrow x \in (-\infty, 1) \cup (2, 3)$$

Thus, $\mathrm{f}(\mathrm{x})$ **is increasing on** $(-\infty, 1) \cup (2, 3)$**. Clearly, it includes answer** (b)**.**

Q.5 Answer:

Both 1 and 2

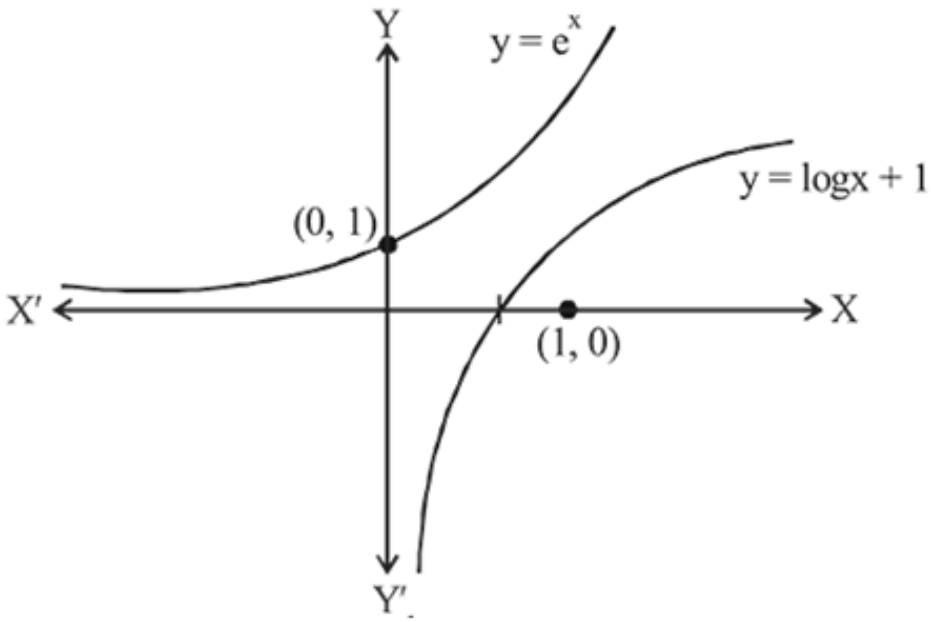
Solution:

$f(x) = \log x$

Clearly $\mathrm{f}(\mathrm{x})$ **is increasing on** $(0, \infty)$

$f(x) = e^x - x\log x$

$f'(x) = e^x - (\log x + 1)$



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From the figure it is clear that $f'(x) > 0$ on $(1, \infty)$.

So both statements $(1) \& (2)$ are correct.

Q.6 Answer:

$(-3, 0)$

Solution:

$f(x) = \sqrt{9 - x^2}$

$f'(x) = \frac{1}{2\sqrt{9-x^2}} \times (-2x) = -\frac{x}{\sqrt{9-x^2}}$

For function to be increasing

$-\frac{x}{\sqrt{9-x^2}} > 0$

or $-x > 0$ or $x < 0$

but $\sqrt{9 - x^2}$ is defined only when $9 - x^2 > 0$ or $x^2 - 9 < 0$

$(x + 3)(x - 3) < 0$

i.e. $-3 < x < 3 \; -3 < x < 3 \cap x < 0 \Rightarrow -3 < x < 0$

Q.7 Answer:

$g\{f(x)\} \leq g\{f(0)\}$

Solution:

$f'(x) > 0$ if $x \geq 0$ and $g'(x) < 0$ if $x \geq 0$ Let $\mathrm{h(x) = f(g(x))}$ then

$h'(x) = f'(g(x)) \cdot g'(x) < 0$ if $x \geq 0$

$\therefore \mathrm{h(x)}$ is decreasing function

$\therefore \mathrm{h(x) \leq h(0)}$ if $\mathrm{x \geq 0}$

$\therefore \mathrm{f(g(x)) \leq f(g(0)) = 0}$

But codomain of each function is $[0, \infty)$ $\therefore \mathrm{f(g(x)) = 0}$ for all $\mathrm{x \geq 0}$ $\therefore \mathrm{f(g(x)) = 0}$

Also $g(f(x)) \leq g(f(0))$ [as above]

Q.8 Answer:

$[\sqrt{2}, \sqrt{10}]$

Solution:

Clearly, domain of the function is $[2, 4]$. Now,

$f'(x) = \frac{1}{\sqrt{x-2}} - \frac{1}{2\sqrt{4-x}}$

$f'(x) = 0$ or $\sqrt{x - 2} = 2\sqrt{4 - x}$

or $x - 2 = 16 - 4x$ or $x = \frac{18}{5}$

Now, $f(2) = \sqrt{2}, f\left(\frac{18}{5}\right) = 2\sqrt{\frac{18}{5} - 2} + \sqrt{4 - \frac{18}{5}} = \sqrt{10}$

$f(4) = 2\sqrt{2}$

Hence, range of the function is $[\sqrt{2}, \sqrt{10}]$. Also, here $x = (18/5)$ is the point of global maxima

Q.9 Answer:

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decreasing in $[0, \infty)$

Solution:

We have $e < \pi$ **and**

$f'(x) = \frac{\frac{1}{\pi+x}\log(e+x) - \frac{1}{e+x}\log(\pi+x)}{\{\log(e+x)\}^2}$

$= \frac{(e+x)\log(e+x) - (\pi+x)\log(\pi+x)}{(\pi+x)(e+x)\{\log(e+x)\}^2}$

In $[0, \infty)$**, denominator** > 0 **and numerator** < 0 **since,**

$e + x < \pi + x$. **Hence,** $f(x)$ **is decreasing in** $[0, \infty)$

Q.10 Answer:

$k > 3$

Solution:

Given $f(x) = kx^3 - 9x^2 + 9x + 3$

On differentiating w.r.t. x**, we get**

$f'(x) = 3kx^2 - 18x + 9$

For a function to be monotonically increasing.

$b^2 - 4ac < 0$

Here , $a = 3k, b = -18, c = 9$

$\therefore \quad b^2 - 4ac = (-18)^2 - 4(3k)(9)$

$= (-18)(-18) - (3k)18 \times 2$

$\Rightarrow \quad 36 - 12k < 0 \quad \Rightarrow \quad k > 3$

Q.11 Answer:

$\left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$

Solution:

$g'(x) = xf'\left(2x^2 - 1\right) - xf'\left(1 - x^2\right)$

$= x\left(f'\left(2x^2 - 1\right) - f'\left(1 - x^2\right)\right)$

$g'(x) > 0$

If $x > 0, 2x^2 - 1 > 1 - x^2$ **(as** f' **is an increasing function)**

or $3x^2 > 2$ **or** $x \in \left(-\infty, -\sqrt{\frac{2}{3}}\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$

or $x \in \left(\sqrt{\frac{2}{3}}, \infty\right)$

or $x \in \left(\sqrt{\frac{2}{3}}, \infty\right)$

If $x < 0, 2x^2 - 1 < 1 - x^2$

or $3x^2 < 2$ **or** $x \in \left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right)$ **or** $x \in \left(-\sqrt{\frac{2}{3}}, 0\right)$

Q.12 Answer:

$a \in$ -1, 5

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Solution:

$f a^2 - 5 > f 4a$

f **is** $\downarrow$

$\Rightarrow a^2 - 5 < 4a$

$\Rightarrow a^2 - 4a - 5 < 0$

$\Rightarrow a - 5a + 1 < 0$

$\Rightarrow -1 < a < 5$

Q.13 Answer:

onto but not one-one

Solution:

$f(x) = 2x^3 - 15x^2 + 36x + 1$

$f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3)$

Thus, $f(x)$ **is increasing in** $[0, 2]$ **and decreasing in** $[2, 3]$**. Therefore** $f(x)$ **is many-one.** $f(0) = 1; \quad f(2) = 29; \quad f(3) = 28$

Range is $[1, 29]$.

Hence, $f(x)$ **is many-one-onto.**

Q.14 Answer:

increases in $-\infty, \infty$

Solution:

$f' x = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}} \geq 0 \forall x \in R$

Q.15 Answer:

$(-1/2, 0) \cup (1/2, \infty)$

Solution:

$f'(x) = 4x - 1/x$

$\mathrm{f}'(\mathrm{x})$ **is monotonic increasing when** $\mathrm{f}'(\mathrm{x}) > 0$

$\Rightarrow 4\mathrm{x} - 1/\mathrm{x} > 0$

$\Rightarrow \frac{4x^2-1}{x} > 0$

$\Rightarrow \begin{cases} 4x^2 - 1 > 0 & \text{when} \quad x > 0 \\ 4x^2 - 1 < 0 & \text{when } x < 0 \end{cases}$

But $x > 0, 4x^2 - 1 > 0 \quad \Rightarrow x^2 > 1/4 \Rightarrow |x| > 1/2$

$\Rightarrow x \in (1/2, \infty)$

and $x < 0, 4x^2 - 1 < 0 \quad \Rightarrow x^2 < 1/4 \Rightarrow |x| < 1/2$

$\Rightarrow x \in (-1/2, 0)$

$\therefore \mathrm{x} \in (-1/2, 0) \cup (1/2, \infty)$

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Q.16 Answer:

$(0,1) \cup (1,4)$

Solution:

$\mathrm{f(x)} = \left(\mathrm{a}^2 - 3\mathrm{a} + 2\right)\left(\cos^2 \mathrm{x}/4 - \sin^2 \mathrm{x}/4\right)$

$\Rightarrow \mathrm{f(x)} = (\mathrm{a}-1)(\mathrm{a}-2)\cos \mathrm{x}/2 + (\mathrm{a}-1)\mathrm{x} + \sin 1$

$\Rightarrow \mathrm{f}'(\mathrm{x}) = -\frac{1}{2}(\mathrm{a}-1)(\mathrm{a}-2)\sin\frac{\mathrm{x}}{2} + (\mathrm{a}-1)$

$\Rightarrow \mathrm{f}'(\mathrm{x}) = (\mathrm{a}-1)\left[1 - \frac{(\mathrm{a}-2)}{2}\sin\frac{\mathrm{x}}{2}\right]$

If $\mathrm{f(x)}$ **does not possess critical points, then** $f'(x) \neq 0$ **for any** $x \in \mathbf{R}$

$\Rightarrow (\mathrm{a}-1)\left[1 - \frac{(\mathrm{a}-2)}{2}\sin\frac{\mathrm{x}}{2}\right] \neq 0$ **for any** $\mathrm{x} \in \mathbf{R}$

$\Rightarrow a \neq 1$ **and** $1 - \left(\frac{a-2}{2}\right)\sin\frac{x}{2} = 0$

must not have any solution in R**.**

$\Rightarrow a \neq 1$ **and** $\sin\frac{x}{2} = \frac{2}{a-2}$ **is not solvable in** R

$\Rightarrow \mathrm{a} \neq 1$ **and** $\left|\frac{2}{\mathrm{a}-2}\right| > 1 \quad \left[\begin{array}{l} \text{For a} = 2, \mathrm{f(x)} = \mathrm{x} + \sin 1 \\ \therefore \mathrm{f}'(\mathrm{x}) = 1 \neq 0 \end{array}\right]$

$\Rightarrow \mathrm{a} \neq 1$ **and** $|\mathrm{a}-2| < 2$

$\Rightarrow a \neq 1$ **and** $-2 < a - 2 < 2$

$\Rightarrow a \neq 1$ **and** $0 < a < 4 \Rightarrow a \in (0,1) \cup (1,4)$

Q.17 Answer:

$a^2 - 3b + 15 < 0$

Solution:

$f(x) = x^3 + ax^2 + bx + 5\sin^2 x$

$d\left(\frac{f(x)}{dx}\right) = 3x^2 + 2ax + b + 5\sin^2 x = f'(x)$

For increasing function $\mathrm{f}'(\mathrm{x}) > 0$

$3x^2 + 2ax + b + 5\sin 2x > 0$

Then

$3x^2 + 2ax + b - 5 < 0$

And $b^2 - 4ac < 0$

$\Rightarrow 4a^2 - 12(b-5) < 0$

$\Rightarrow a^2 - 3b + 15 < 0$

$\Rightarrow a^2 - 3b + 15 < 0$

Q.18 Answer:

Increasing in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

Solution:

$f'(x) = -12\cos^3 x \sin x - 30\cos^2 x \sin x - 12\cos x \sin x = -6\sin x \cos x(\cos x + 2)(2\cos x + 1)$

$\mathrm{f}'(\mathrm{x}) = 0$, **for** $\mathrm{x} = 0, \frac{\pi}{2}, \frac{2\pi}{3}, \pi$

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Clearly, $f'(x) > 0$ **for** $\frac{\pi}{2} < x < \frac{2\pi}{3}$
And $f'(x) < 0$; **for** $0 < x < \frac{\pi}{2}$ **or** $\frac{2\pi}{3} < x < \pi$

Q.19 Answer:

$[-1, 1]$

Solution:

$f'(x) = 3\left(\frac{a^2-1}{a^2+1}\right)x^2 - 3$
$f'(x) < 0$ **for all** x **if** $a^2 - 1 \leq 0 \Rightarrow -1 \leq a \leq 1$

Q.20 Answer:

3

Solution:

$f'x < 0$
$\therefore fx^3 + fx \leq f\text{-}fx\text{ - }x^3$
$\Rightarrow x^3 + fx \geq \text{ - }fx\text{ - }x^3$
$\Rightarrow fx + x^3 \geq 0$
$\Rightarrow 3x^2 + 6x\text{ - }1 \leq 0$
As $x \in Z, x \in \{\text{ - }2,\text{ - }1, 0\}$

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